4-LABORATORIYA ISHI

Mavzu: Linear Regression yordamida prognoz qilish (Uy narxini bashorat qilish)

Maqsad

Ushbu laboratoriya ishining maqsadi:

- Linear Regression modelini qurish
- Modelni o‘qitish
- Bashorat qilish
- MSE va R^2 hisoblash
- Natijani grafik orqali tahlil qilish

Kerakli vositalar

- Python 3.10+
- Pandas
- Matplotlib
- Scikit-learn

O‘rnatish:

```
pip install pandas matplotlib scikit-learn
```

Nazariy asos

Linear Regression — chiziqli bog‘liqlikni aniqlovchi model.

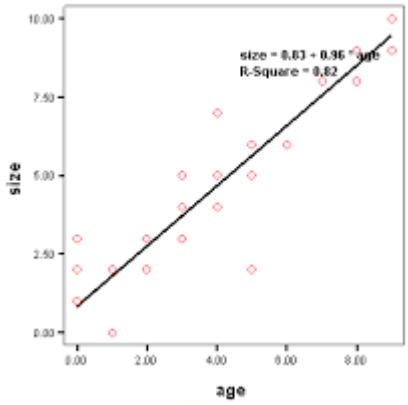
Formula:

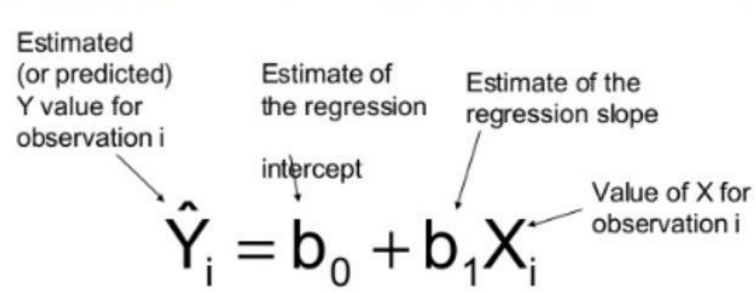
y=b0+b1xy = b_0 + b_1 xy=b0+b1x

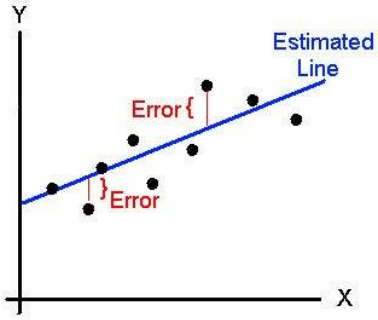
Bu yerda:

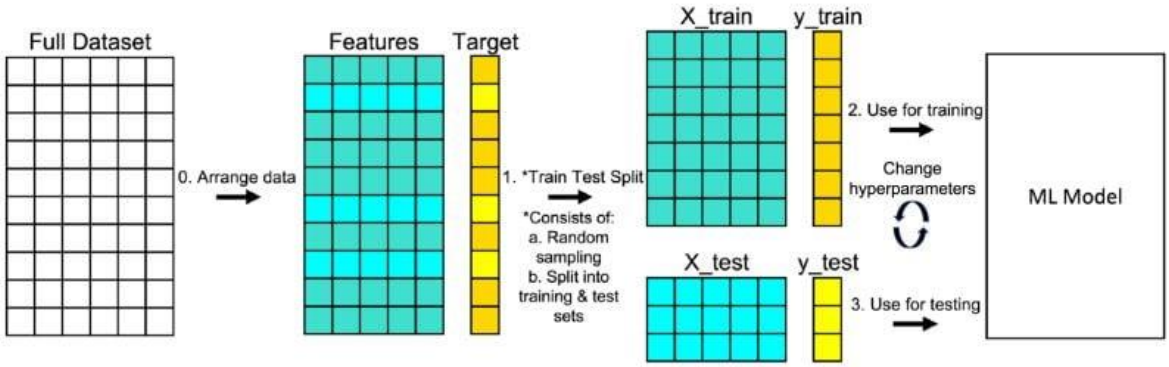
- yyy — natija (uy narxi)
- xxx — kiruvchi parametr (maydon)
- b0b_0b0 — intercept
- b1b_1b1 — koeffitsient

Linear regression ishlash jarayoni









Ishlash tartibi (Step-by-Step)

1-qadam: Dataset yaratish

students emas, yangi dataset (house.csv):

```
area,price
50,50000
60,60000
80,80000
100,100000
120,120000
150,150000
200,200000
```

2-qadam: Dataset yuklash

```
import pandas as pd

df = pd.read_csv("house.csv")
print(df.head())
```

3-qadam: X va y ajratish

```
X = df[["area"]]
y = df["price"]
```

4-qadam: Train-test ajratish

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

5-qadam: Model yaratish va o‘qitish

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)
```

6-qadam: Bashorat qilish

```
pred = model.predict(X_test)
print("Bashorat:", pred)
```

7-qadam: Modelni baholash

```
from sklearn.metrics import mean_squared_error, r2_score

mse = mean_squared_error(y_test, pred)
r2 = r2_score(y_test, pred)

print("MSE:", mse)
print("R2:", r2)
```

8-qadam: Grafik chizish

```
import matplotlib.pyplot as plt

plt.scatter(X, y)
plt.plot(X_test, pred, color="red")
```

```
plt.xlabel("Area")
plt.ylabel("Price")
plt.title("Linear Regression")
plt.show()
```

To‘liq kod (Yakuniy variant)

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

df = pd.read_csv("house.csv")

X = df[["area"]]
y = df["price"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

pred = model.predict(X_test)

mse = mean_squared_error(y_test, pred)
r2 = r2_score(y_test, pred)

print("MSE:", mse)
print("R2:", r2)

plt.scatter(X, y)
plt.plot(X_test, pred, color="red")
plt.xlabel("Area")
plt.ylabel("Price")
plt.title("Linear Regression")
plt.show()
```

Talaba tekshiradi:

✅ Model o‘qitildi
✅ Bashorat chiqdi
✅ MSE va R^2 hisoblandi
✅ Grafikda regression chizig‘i chiqdi

Hisobot topshirish talabi

Talaba topshiradi:

1. Python kodi

2. Grafik skrinshoti
3. 8-10 qatordan iborat tahlil:
    - R^2 nimani bildiradi?
    - MSE qancha chiqdi?
    - Model qanchalik aniqlikda ishladi?

Mezoni	**Ball**
Model ishlashi	40
Baholash ko‘rsatkichlari	25
Grafik to‘g‘riligi	15
Tahlil yozish	20